Qingyang Liu

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Personal Website: https://xxxco.github.io

Education

Duke University, School of Medicine, Durham, NC — *Aug. 2024 – May 2026*
Master of Biostatistics

University of Washington, Seattle, WA — *Sept. 2020 – June 2024*
BS in Applied Computational Mathematical Science

Experience

Machine Learning Research Assistant, **Duke University Rohit Singh Lab** — *Oct. 2024 - Present*

- Designed a topology-based network selection algorithm using degree scoring across 33 biological networks
- Built DE re-ranking pipeline integrating scRNA-seq signals with network rankings, improving precision @ 10 by 100%
- Reduced pipeline runtime by 4x enabling scalable rare disease gene prioritization at research scale
- Launched a Streamlit ML app enabling user input, real-time inference, and decision-support outputs

Data Science Research Assistant, Duke Center for AIDS Research — *Oct. 2024 - Present*

- Built R pipeline evaluating HIV transmission clustering across 7 genomic subregion and 90+ parameters sets
- Benchmarked ClusterPicker/HIV-TRACE across distance and bootstrap thresholds on 584 sequences
- Identified concordance and clustering yield as distinct performance dimensions via V-measure analysis
- Visualized multi-region clustering comparisons across methods and thresholds using ggplot2

Algorithm Engineering Intern, Ping'an Technology — *July 2023 – Sept. 2023*

- Applied interpolation and curve fitting to improve synchronization between speech and lip movements in AI figures
- Automated analysis of 80,000+ videos in Python, extracting metadata and streamlining workflows for efficiency
- Built and managed databases by uploading and classifying 11,000+ minutes of video data for AI training pipelines
- Processed and optimized raw video data with FFmpeg, adjusting parameters to maximize clarity and visual quality

Projects

Pilot Trial Designing Software, Duke University — *Aug. 2024 - Present*

- Built simulation and Bayesian models to assess pilot-trial feasibility, dropout risk, and data-driven design decisions
- Developed a Shiny app for automated Bayesian scenario testing across sample sizes and prior assumptions

LLM Safety & Interpretability with Sparse Autoencoders, Duke University — *Nov. 2025 – Dec. 2025*

- Identified toxicity- and hallucination-aligned SAE features and built classifiers with strong detection performance
- Applied feature-level steering to hidden states to reduce unsafe generations and improve answer accuracy

Reinforcement Learning Algorithm Benchmarking, Duke University — *Nov. 2025 – Dec. 2025*

- Benchmarked DQN, A2C, and PPO with ablations to assess key algorithm components across control tasks
- Analyzed stability and efficiency, highlighting actor-critic advantages in continuous-control environments

Vaccine Reservation System, University of Washington — *Dec. 2021*

- Designed a relational SQL database with Java integration, including queries, procedures, and supply-chain simulation
- Built user-facing modules for secure login, appointment scheduling, and availability monitoring for vaccine reservation

Skills

Programming Language: Python, R, SQL, Java
Machine Learning: PyTorch, Scikit-learn, LightGBM, TensorFlow, XGBoots, Cross-Validation
Data Analysis: A/B Testing, Statistical Modeling, Pandas, NumPy, Matplotlib, Seaborn, Shiny, Data Visualization
Tools & Platforms: Git, Docker, FFmpeg, LaTeX, Linux